



SUPPLY CHAIN INTELLIGENCE // FREE CASE STUDY // DEFENSE ELECTRONICS // EU (GERMANY)

HENSOLDT

Europe's AESA Radar Champion Has No European GaN Supply Chain

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Hensoldt is Germany's leading defence electronics company — AESA radar for the Eurofighter, passive radar (Twinvis), electronic warfare, and optronics. It is the sensor backbone of European air defence. Every AESA radar it builds depends on gallium nitride Transmit/Receive modules — the GaN amplifiers that make phased-array radar possible. Europe has no domestic GaN wafer fabrication at defence-relevant quality. Hensoldt's GaN supply chain runs through Wolfspeed (US) and United Monolithic Semiconductors (France/Germany) — both constrained by the same gallium price spike that is hitting every defence electronics company in the world. This case study maps what that means for Eurofighter deliveries and the FCAS next-generation fighter programme.

\$687/kg

GALLIUM SPOT PRICE

UP 204% SINCE
PRE-EXPORT-CONTROL
BASELINE**ZERO**

EUROPEAN DOMESTIC GALLIUM

PRIMARY PRODUCTION
CAPACITY**98%**

CHINA SHARE OF GLOBAL

LOW-PURITY GALLIUM
PRODUCTION**FCAS**

FUTURE COMBAT AIR SYSTEM

NEXT-GEN PROGRAMME AT RISK

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INTELLIGENCE ALERT // HENSOLDT // THE GaN SUPPLY CHAIN GAP

THE CORE FINDING

Hensoldt builds the CAPTOR-E AESA radar for the Eurofighter Typhoon — the air defence backbone of Germany, Spain, the UK, Saudi Arabia, and Qatar. AESA radar is defined by its GaN Transmit/Receive modules. Each CAPTOR-E radar contains hundreds of GaN T/R modules. Each T/R module uses GaN wafers fabricated from gallium metal. China controls 98% of global gallium production. China has applied export controls that have driven gallium to \$687/kg. Europe has zero domestic gallium production. Hensoldt's AESA radar capability — the capability that protects European airspace — depends on a supply chain it does not control.

THE EUROFIGHTER EXPOSURE // CAPTOR-E AESA — GaN T/R MODULES — CHINESE GALLIUM

The CAPTOR-E AESA radar replaced the CAPTOR-M on Eurofighter Typhoon from Tranche 3B onward. Hensoldt is the prime contractor. The radar's AESA capability depends on an Active Electronically Scanned Array of GaN T/R modules — each module a GaN power amplifier and low-noise amplifier pair. CAPTOR-E has approximately 1,400 T/R modules per radar. At typical GaN content of 0.3–0.5g gallium per module, each CAPTOR-E radar contains approximately 420–700g of refined gallium. At \$687/kg, the gallium content per radar is approximately \$290–\$480. That sounds manageable — until you trace where the gallium comes from.

SUPPLY CHAIN TIER	ENTITY	GEOGRAPHY	VULNERABILITY	HENSOLDT EXPOSURE
Gallium feedstock (Tier 3)	Guangdong province smelters	PRC — 98% global share	Export controls active since August 2023 — tightening Q3 2026	CRITICAL — no European alternative at scale
GaN wafer fabrication (Tier 2)	Wolfspeed, Qorvo, MACOM (US); UMS (France/Germany)	US and France/Germany	Both constrained by commercial 5G demand; gallium price spike	HIGH — lead times extending; price increasing
GaN T/R module assembly (Tier 1)	UMS (Orsay, France) — primary EU source	France — partial EU alternative	EU-based — reduces ITAR risk	MEDIUM — capacity constrained but EU-sovereign
CAPTOR-E radar integration	Hensoldt Ulm, Germany	Germany	Domestic — no geographic risk	LOW — fully controlled
Eurofighter platform	Airbus Defence & Space; BAE Systems; Leonardo	Germany/Spain/U K/Italy	Multi-national but all NATO	LOW — platform sovereign

THE FCAS RISK // NEXT-GENERATION FIGHTER — GaN AT 3X THE SCALE

The Future Combat Air System (FCAS) — the joint Franco-German-Spanish sixth-generation fighter programme — will use an AESA radar substantially more capable than CAPTOR-E. Each FCAS airframe is expected to carry multiple AESA apertures: forward, side-looking, and tail. The combined GaN T/R module count per FCAS aircraft is estimated at 3,000–5,000 modules — 2–3 times the Eurofighter. At FCAS production volumes of 200–300 aircraft across Germany, France, and Spain, the GaN demand represents a strategic material acquisition of significant scale. There is no European gallium primary production. There is no European GaN wafer fab at defence-relevant volume. The FCAS radar supply chain is being designed on top of a dependency that neither France, Germany, nor Spain has a plan to close.

THE STRATEGIC IRONY

Hensoldt is the company that makes European air defence credible — and its most critical input is sourced from the country against which European air defence may one day need to operate. This is not a Hensoldt management failure. It is a European industrial policy failure. But Hensoldt carries the programme risk.

WHAT A FULL ENGAGEMENT WOULD ADD

A full Defense.Codes engagement would add: (1) Hensoldt GaN T/R module sourcing profile — UMS vs Wolfspeed vs MACOM split; (2) CAPTOR-E production schedule vs gallium availability model; (3) FCAS GaN demand projection 2030–2040; (4) European gallium recovery programme assessment from French aluminium byproduct; (5) Competitor analysis — Leonardo radar division vs Thales vs Indra on GaN sourcing.

This case study is one of five free public intelligence samples published by Defense.Codes. Category: DEFENSE ELECTRONICS — EU. Full-depth engagement reports available for purchase. Contact: ceo@capa.cloud | defense.codes

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Sources: Hensoldt AG annual reports 2023–2025 — Hensoldt CAPTOR-E technical publications — Eurofighter Typhoon programme documentation — FCAS programme statements (DGAP) — UMS (United Monolithic Semiconductors) product disclosures — Rotterdam Exchange gallium pricing Q2 2026 — China MOFCOM export control notices — CSIS critical minerals analysis July 2025 — Wolfspeed investor relations Q2 2026.

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